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Effects of a 12-week gait retraining program on the Achilles tendon adaptation of habitually shod runners

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Abstract

Purpose: This study investigated the effects of a 12-week gait retraining program on the morphological and mechanical properties of the Achilles tendon (AT) during running on the basis of real-time dynamic ultrasound imaging.

Methods: A total of 30 male recreational runners who were used to wearing cushioned shoes with a rearfoot strike (RFS) pattern were recruited. They were randomized into a retraining group (RG, $n=15$) and a control group (CG, $n=15$). The RG group was asked to run in five-fingered minimalist shoes with a forefoot strike (FFS) pattern, and the CG group was asked to keep their strike pattern. Three training sessions were performed per week. All the participants in RG uploaded running tracks obtained through a mobile application (.jpg) after each session for training supervision. The ground reaction force, kinematics, and kinetics of the ankle joint at 10 km/h were collected using an instrumented split-belt treadmill and a motion capture system. The morphological (length and cross-sectional area) and mechanical characteristics (force, stress, strain, etc.) of AT in vivo were recorded and calculated with a synchronous ultrasonic imaging instrument before and after the intervention. Repeated two-way ANOVA was used to compare the aforementioned parameters.

Results: A total of 28 participants completed the training. The strike angle of RG after training was significantly smaller than that before training and significantly smaller than that of CG after training ($F_{(1, 13)}=23.068$, $p<0.001$, partial $\eta^2=0.640$). The length ($F_{(1, 13)}=10.086$, $p=0.007$, partial $\eta^2=0.437$) and CSA ($F_{(1, 13)}=7.475$, $p=0.017$, partial $\eta^2=0.365$) of AT in RG increased after training. A significant main effect for time was observed for the time-to-peak AT force ($F_{(1, 13)}=5.225$, $p=0.040$, partial $\eta^2=0.287$), average ($F_{(1, 13)}=7.228$, $p=0.019$, partial $\eta^2=0.357$), and peak AT loading rate ($F_{(1, 13)}=11.687$, $p=0.005$, partial $\eta^2=0.473$).

Conclusion: Preliminary evidence indicated that a 12-week gait retraining program could exert a beneficial effect on AT. 57% (8/14) runners in RG shifted from RFS to FFS pattern. Although not all runners were categorized as FFS pattern after the intervention, their foot strike angle was reduced. Retraining primarily positively promoted AT morphological properties (i.e., CSA and length) to strengthen AT capability for mechanical loading.

KEYWORDS
Achilles tendon force, loading, running, strain, strike pattern

1 | INTRODUCTION

As one of the strongest tendons in the human body, the Achilles tendon (AT) plays a crucial role in movements, such as walking, running, and jumping, by transmitting the forces of the triceps muscle in the lower leg.[1] However, a recent systematic review reported that Achilles tendinopathy (13.7%) accounted for one of the highest incidence proportions of running-related injuries.[2] Although Achilles tendinopathy progresses slowly, the injury can worsen if neglected, and full recovery can take a year or more.[3] Stimulating AT within appropriate limits has been proved to result in positive (anabolic) adaptation. Therefore, based on the model of tendon pathology, improving the morphological and mechanical properties of AT and preventing the occurrence of early "minor" symptoms are of considerable economic value and social significance.

On the basis of the differences in foot strike patterns during running, runners who habitually wore minimalist shoes with a forefoot strike (FFS) pattern had greater AT cross-sectional area (CSA), stiffness, and Young's modulus than runners who habitually wore conventional shoes with a rearfoot strike (RFS) pattern, indicating an advantageous adaptation for endurance runners.[4–6] Compared with RFS, FFS requires greater activation of the triceps surae to control the heel descent during the eccentric phase. The plantar-flexion force produced by the contraction of the triceps surae is transmitted through AT, combined with a higher stretching speed, resulting in increased mechanical stress on AT.[7] As a mechanoresponsive structure, the mechanical properties of AT can be improved by exerting an increased load. In general, progressive exercises that provide an appropriate mechanical loading of AT are the most recommended strategy for improving AT function.[8,9] Statistically, more than 95% of traditionally shod runners adopt an RFS pattern when they run on modern hard surfaces.[10] Therefore, it was reasonable to propose a gait retraining program that converted the most common RFS pattern to the FFS pattern. Nevertheless, when we are accustomed to wearing cushioned running shoes, transitioning to an FFS pattern without proper preparation involves risk. One study reported 10 cases of foot injuries (metatarsal stress fracture, heel stress fracture, and plantar fasciitis) associated with abrupt transitions. The average time for these 10 cases to convert to running in minimalist shoes for the entire time was 3 weeks.[11] In particular, half of the runners converted to the minimalist technique completely without a transition period. Therefore, it is vital to convert to an FFS pattern in minimal shoes gradually, as the musculoskeletal system needs sufficient time to adapt to the changes in loading. However, Fuller et al.[12] reported after a 6-week transition to minimalist footwear, there were no further effects on performance, running economy, or altered running biomechanics and lower limb bone mineral density after 20 weeks of follow-up. Similarly, Ekizos et al.[13] observed no significant changes in the oxygen consumption and rate of metabolic energy consumption after 14 weeks of intervention from RFS to FFS. Considering that shorter training sessions may cause additional foot/ankle injury,[11] while longer training sessions may not further improve AT properties,[12,13] the current study proposed 12 weeks of progressive gait retraining, i.e., gradually transitioning from RFS to FFS by wearing minimalist shoes.[14,15]

Notably, accurately and effectively obtaining the biomechanical characteristics of AT is a challenging task. As a highly reliable method, ultrasound imaging is comprehensively used at present to assess tendon morphology (tendon length and CSA) in vivo under static or quasi-static conditions.[16,17] However, only a few studies have applied it to the real-time measurement of AT length in vivo during running. Specifically, Waugh et al.[18] used an ultrasound probe to directly measure muscle and tendon excursion between children and adults performing vertical hopping. Suzuki et al.[19] directly measured the entire length of the AT during forefoot and rearfoot running at different running speeds on a treadmill. Nevertheless, since these studies were conducted under barefoot conditions and the limitations of the experimental conditions (inability to measure force, low accuracy of kinematic measurements, etc.), their representation of actual human motion might be limited. Therefore, on this basis, we optimized a direct measurement method for AT in vivo under shoe-wearing conditions, namely, an ultrasound probe was fixed at the muscle–tendon junction of AT and gastrocnemius medialis (MTJ_{GM_AT}) through an elastic bandage. Then, a 3D force-measuring platform was synchronized to evaluate the morphological and mechanical characteristics of AT during running.

The current study aimed to determine changes in the morphological and mechanical properties of runners' AT after a progressive 12-week foot strike gait retraining program that converts to an FFS pattern in minimal shoes. We hypothesized that after the 12-week retraining program: (1) the majority of runners who were accustomed to RFS would transition to FFS; (2) the morphological properties (length and CSA) of AT would increase; and (3) the peak force, stress, corresponding loading rate, and elongation of AT would increase.

2 | METHODS

2.1 | Participants

On the basis of previous data published by Joseph et al.[20] (changes in AT stiffness before and after gait retraining, $f=0.26$), a priori power analysis was conducted for expected outcomes with a Type I error probability of 0.05 and a power of 0.8. This analysis indicated that a sample size of 26 would pro. This study was performed from April 6, 2021, to August 21, 2021. Considering the 15%–20% dropout rate, a total of 33 male recreational runners were recruited from collegiate, and local clubs were screened, but three were excluded (Figure 1). Thirty male recreational runners were randomly assigned by random sequence generation to either a retraining group (RG, $n=15$, age: 34.53 ± 8.04 years, height: 173.33 ± 6.49 cm, weight: 70.18 ± 8.06 kg) or a control group (CG, $n=15$, age: 32.06 ± 10.19 years, height: 173.10 ± 5.22 cm, weight: 67.53 ± 9.45 kg). In this study, the random grouping method was used as a random sequence generation method. Using an online random number generator, a range of random numbers from 1 to 30 was determined, and a sequence of random numbers was generated. Each random number represented an individual participant. Participants were allocated to different groups in the order of the generated random sequence. Specifically, participants with odd-numbered sequence values (1, 3, 5, etc.) were assigned to RG, while those with even-numbered sequence values (2, 4, 6, etc.) were assigned to CG. The inclusion criteria were as follows: (1) habitual rearfoot strikers who wore cushioned shoes without experience in minimalist shod running; (2) running mileage was more than 20 km per week in the past 4 weeks and intending to continue at a similar intensity for the following 12 weeks; and (3) no neurological conditions or lower limb injuries in the last 3 months, and not currently undergoing physical therapy or lower limb rehabilitation. No caffeine/alcohol and no strenuous exercise had been consumed/performed the last 2/24 h prior to the measurements. All the participants provided with written informed consent approved by the ethics committee of Shanghai University of Sport (No. 102772021RT085).

2.2 | Procedure

An ultrasound device (uSmart 3300, Terason) was used to assess AT length in a relaxed state. The participants were asked to lie prone on a treatment bed with their dominant ankle in a neutral position (90°).[7] An ultrasound gel was applied to the head of a 12L5A probe (maximum frequency:

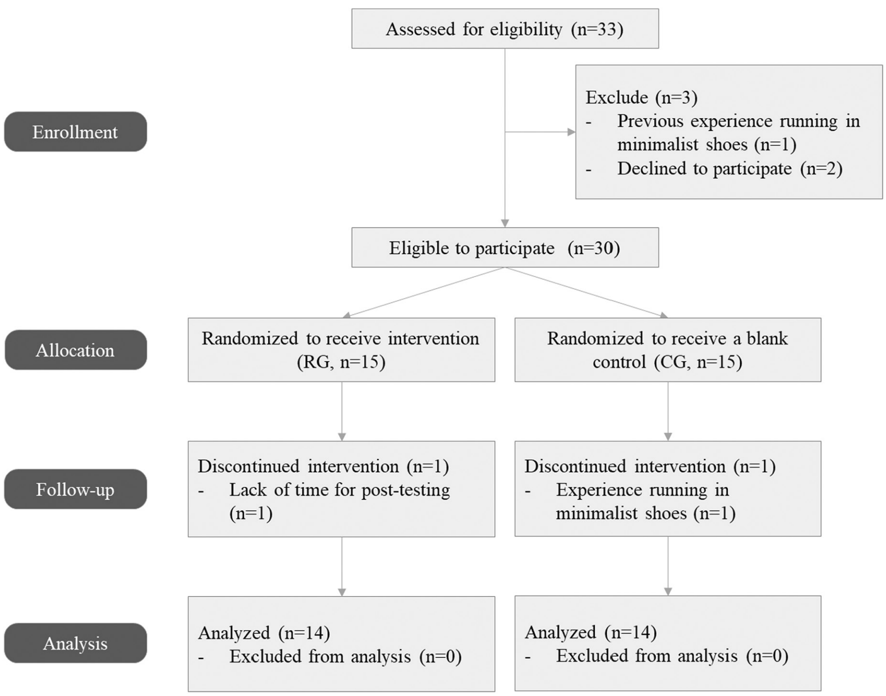

FIGURE 1 CONSORT flowchart of the enrollment and follow-up. CG, control group; RG, retraining group.

12 MHz; probe array length: 4.5 cm) at MTJ_{GM_AT}. A 25-gauge needle was placed between the ultrasound probe and the skin surface to identify MTJ_{GM_AT} in the ultrasound imaging with the probe oriented on the sagittal plane. The corresponding labeling where the needle and the ultrasound probe crossed was marked on the skin. The same manipulation was performed to mark the insertion point of AT on the calcaneus. The length of the two points was measured as the resting length of AT by using tape. Then, the probe was placed at the lateral and medial malleoli perpendicular to AT to collect transverse images of CSA.[17] In the preliminary experiment, the intraclass correlation coefficient (ICC) was used to evaluate the reliability of the measurement results of different experimenters and the same experimenter at various time points. The results showed that the morphological properties of AT collected and calculated via ultrasound technology achieved good reliability (ICC = 0.895–0.996).

Before running trials, each participant was fitted with exercise clothing and neutral cushioned running shoes (Nike Air Zoom Pegasus 34, weight: 285 g, size: 43, heel–toe drop: 12 mm, foam material, and zoom air cushion insole) provided by the experimenters. The participants warmed up for 5 min at a self-selected running speed on an instrumented treadmill (Bertec FIT, Bertec). Then, 36 infrared retroreflective markers with a diameter of 14.0 mm were bilaterally attached onto the lower extremities to define the hip, knee, and ankle joints in accordance with a plug-in gait marker set. Notably, we made a hole in the heel cup of the shoe because one of the markers must be placed on the insertion point of AT on the calcaneus. This point was identified via ultrasound when measuring the resting length of AT. The ultrasonic probe applied to the gel was fixed to MTJ_{GM_AT} on the dominant side through a homemade foam fixture and elastic bandages. Then, the sagittal length changes of the AT were recorded via a dynamic ultrasound video at 25 Hz (Figure 2). Four reflective markers were also placed on the surface of the ultrasound probe to determine the position of the probe in the laboratory coordinate system. After collecting the static model, the subjects were required to run on the instrumented treadmill at a speed of 10 km/h with their preferred foot strike patterns. The trajectory of the markers was collected with a 10-camera infrared 3D motion capture system (Vicon T40, Oxford Metrics) at a sampling rate of 200 Hz. The ground reaction force (GRF) was measured at a sampling rate of 1000 Hz. The kinematic data, GRF, and ultrasound videos of AT were collected simultaneously with the Vicon system for 10 s after the participants adapted to the running speed. All the participants had sufficient time to adapt to treadmill running with markers, and the probe was attached before data acquisition. The participants were not aware when data were being captured. During the whole experiment, the experimenters were forbidden to remind the participants to pay attention to their foot strike pattern through language or other means.

2.3 | Intervention

Each participant in RG was given a pair of minimalist shoes (Vibram FiveFingers, weight: 139 g, size: 43, heel–toe drop: 0 mm, 3 mm rubber outsole, and no insole) for a 12-week gait retraining program. During the training intervention, the participants were required to run with the FiveFingers shoes at a self-selected speed. The participants were instructed to run with FFS; that is, they were required to run with the ball of the foot at initial contact followed by contact with the rest of the foot. The instructions for FFS before intervention were presented by national first-class long-distance runners who were used to running with FFS. The training plan was prepared by the experimenter in accordance with the individual weekly mileage of each participant after the pretest. The participants must complete a run three times per week with the minimalist shoes for the 12-week program.[20] For Weeks 1 and 2, the participants ran 10% of their weekly mileage in minimalist shoes with FFS and the remaining 90% of the running distance in their usual cushioned running shoes with their preferred foot strike pattern. For Weeks 3–11, the amount of running distance in minimalist shoes was increased by 10% per week; that is, the participants ran 20% of their weekly mileage in minimalist shoes for Week 3, 30% of their weekly mileage in minimalist shoes for Week 4, and so on. Until Week 11, the participants were required to wear minimalist shoes with FFS within a run, i.e., run 100% of their individual weekly mileage in minimalist shoes with FFS for Week 11. Full minimalist shoe and FFS running were maintained in week 12. In addition, participants were instructed to perform foot core exercises before and after each training session in accordance with the recommendations of clinical rehabilitation physicians to enable the lower extremity musculoskeletal and soft tissue systems to adapt to different foot strike patterns and high loading for a long time. Foot core exercises including double leg heel raises (level surface), double leg heel raises (on step), single leg heel raises (level surface), towel curls, toe spread and toe squeeze, and doming performed for 20 min per time.[21] All the participants uploaded screenshots of their running tracks obtained through a mobile application (.jpg) in the online communication groups, which consisted of both participants and supervisory personnel, after each training session. All participants were required to self-report any discomfort or sensations they experienced during the training sessions allowing us to capture the individualized perceptions of

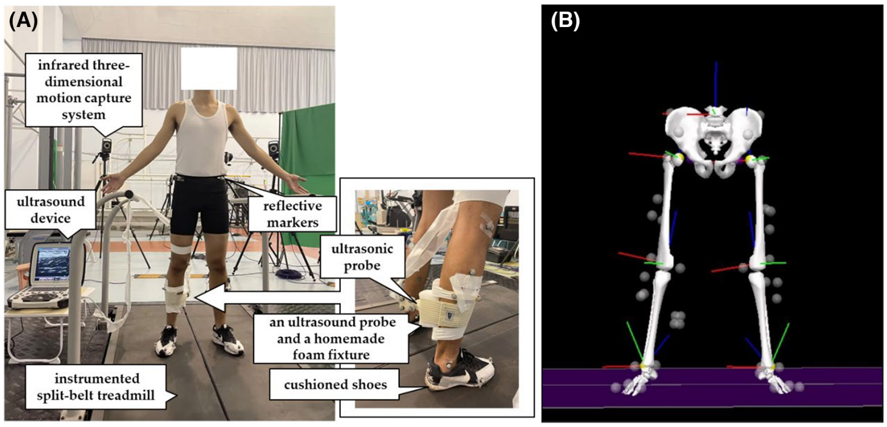

FIGURE 2 (A) Experimental setup; (B) marker model in V3D software.

discomfort directly from the participants in time. Each week, the experimenter and an elite runner went to the field where the participants typically run to correct and guide the foot strike patterns of the participants with the use of slow-motion videos (iPhone 12 Pro Max iOS version 15.1.1, 240 Hz, 1080p resolution, Apple Inc.) for their observation. The participants in CG continued to run in their usual cushioned running shoes with their preferred foot strike patterns for 12 weeks. During the 12-week gait retraining program, all the participants were not allowed to run weekly more than they did before the start of the experiment and were prohibited from participating in any related events.

2.4 | Data processing

The 3D coordinates of the reflective markers were filtered through a Butterworth fourth-order, low-pass filter at a cutoff frequency of 7 Hz with V3D software (v5, C-Motion, Inc.). The analyzed stance phase was defined as the time interval from the initial contact to toe-off the ground. The initial contact was defined as the moment at which GRF exceeded 30 N.[22] Five stance phases in which the ultrasound image data sequence was clearly distinguished were extracted for data analysis.

The model we used in the V3D software was a multi-rigid-body six-degree-of-freedom connection model based on the Hanavan anthropometric model (Figure 2). Hip, knee, and ankle joint centers were calculated by the coordinates from a static calibration trial with local coordinate systems for each segment in participants. The hip joint center was defined as 25% of the distance from the ipsilateral to the contralateral greater trochanter marker. The knee joint center was located midway of a line between the lateral and medial femoral condyles markers. The ankle joint center was located midway of a line between the lateral and medial malleoli markers. The coordinate position of the end of a human body segment can be calculated by a motion capture system using the laboratory coordinates of the marker points on the surface of the segment. The tracking of the segment coordinate system inside V3D performed the details of these calculations automatically. Cadence was defined as the times from the foot strike to ipsilateral foot strike per minute. The strike angle was obtained by subtracting the angle between the foot and the ground at contact from the angle between the foot and the ground while standing. In particular, RFS was determined as a strike angle of >8.0°, while FFS was determined as a strike angle of <1.6°.[22] The Euler angles between the local coordinates of the foot and shank segments were calculated to obtain the ankle contact angle, the peak plantar-flexion/dorsiflexion angle, and the ankle joint range of motion during the stance phase. The peak ankle joint moment was calculated using an inverse dynamic approach.

ImageJ software (version 1.46r, NIH) was used to evaluate the CSA images of AT measured by tracing the surrounding echogenic boundary of AT. The changes in MTJ_{GM_AT} during running were semiautomatically tracked using Tracker video modeling software (Tracker 4.95, InstallBuilder). The position of MTJ_{GM_AT} was tracked automatically on the basis of the in-house scripts of Tracker video modeling. Then, the data were checked manually frame by frame, and the offset points were moved to the correct position. The aforementioned process was repeated three times to reduce the error, and the average values were taken. After the manually checked data were acquired, linear interpolation was applied to adjust the sampling rate of the ultrasound image data sequences. The sampling rate was then changed to 200 Hz, which was consistent with the sampling frequency of the kinematic data. For AT length calculation, the position of MTJ_{GM_AT} in the ultrasound image was mapped into the global coordinate system on the basis of the global coordinates of the four markers on the ultrasound probe. AT length was calculated as the linear distance between MTJ_{GM_AT} and the calcaneus marker coordinates by using

	Retraining group ($n=14$)	Control group ($n=14$)	*p* Value
Age (year)	34.53 ± 8.34	32.60 ± 10.19	0.624
Height (cm)	173.33 ± 6.73	173.10 ± 5.22	0.939
Body mass (kg)	70.18 ± 8.33	67.53 ± 9.45	0.395
Running distance (km/week)	39.00 ± 14.26	39.73 ± 23.81	0.813
Running experience (year)	4.93 ± 2.13	4.93 ± 2.60	0.938

TABLE 1 Characteristics of included participants (mean ± SD).

a custom-written program (MATLAB, Mathworks). Since the motion capture system returns the 3D spatial coordinates of the marker's centroid (14 mm in diameter, plus 3 mm of marker support) and not the actual position on the subject's skin, to calculate correctly the insertion point of AT, a distance of 10 mm (radius = 14/2 mm + 3 mm) was subtracted from the actual marker coordinates.[23] The peak AT strain was calculated as the percentage of the ratio of the maximum length change of AT to the resting length of AT. The peak AT force was calculated by deriving from the ankle joint moment and moment arm of AT, where the value of the moment arm was estimated using an estimation equation based on the polynomial algorithm and AT in vivo imaging data of Lyght et al.[7] and Rugg et al.[24] The detailed evaluation methods could be found in our previous study.[25] In the AT force–time curve, the peak and average loading rate of AT force were determined via the instantaneous and average slope from the initial contact to the peak force, respectively. AT stress was calculated by dividing AT force by the CSA of each participant. AT force impulse during running was calculated by integrating the AT force–time curve by applying the trapezoid rule. AT work was obtained from the area under the descending limb of the AT force–elongation curve. AT's kinetics data and mechanical variables were normalized to body weight.

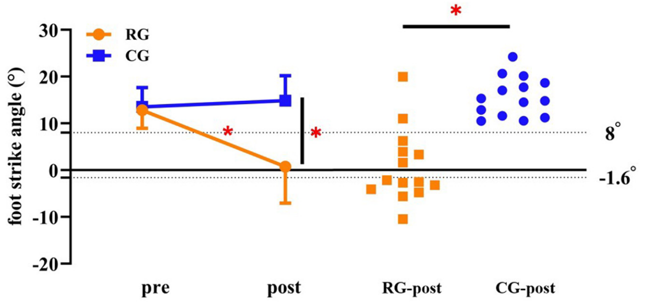

FIGURE 3 Effects of the 12-week gait retraining on foot strike angle at initial contact. CG, control group; RG, retraining group; rearfoot strike pattern: foot strike angle >8°; forefoot strike pattern: foot strike angle <−1.6°; *$p<0.05$: significantly different between the two groups after the 12-week training.

2.5 | Statistical analysis

All data expressed as mean ± standard deviation was analyzed using SPSS statistical software (IBM SPSS v22.0, IBM Corp.). Normality was assessed with Shapiro–Wilk tests. Independent sample *t*-tests were used to compare the basic sociodemographic data of the two groups. A 2 × 2 repeated measure ANOVA was used to compare the differences in the ankle joint kinematics and kinetics, GRF, and morphological and mechanical properties of AT within and between the two groups (RG versus CG) over two time points (pre-intervention versus post-intervention). The effect size (partial η^2) of each parameter in this study was calculated. A significant interaction was detected in ANOVA, and differences across groups or time were quantified using Bonferroni's post hoc multiple comparison test. The significance level was set at the conventional 0.05 level.

3 | RESULTS

All dependent variables were distributed normally as indicated by the Shapiro–Wilk tests. Among the 30 recruited participants, 28 participants (14 in RG and 14 in CG) completed the 12-week protocol and were included in the final analysis (Table 1). During the intervention, one participant in CG ran in minimalist shoes during the training, and 1 participant in RG was unable to participate in the posttest due to a business trip. Those data were excluded (Figure 1). There were no discomforts reported by the participants in the RG group and no running-related injury events occurred during the conduct of the study. The total completion rate of the 12-week gait retraining program was 93.3%.

A significant group-by-time interaction was observed for strike angle with a large effect size in terms of variance explained ($F_{(1, 13)}=23.068$, $p<0.001$, partial $\eta^2=0.640$). Post hoc analyses revealed that the strike angle of RG after training was significantly smaller than that before training and significantly smaller than that of CG after training. Of the 14 participants in RG, eight participants transitioned to FFS (i.e., strike angle <−1.6°), and the transition rate was 57.1% (Figure 3).

For the kinematics and kinetics at the ankle, significant group-by-time interactions with large effect sizes were observed for the ankle angle at foot strike ($F_{(1, 13)}=14.927$, $p=0.002$, partial $\eta^2=0.535$), peak dorsiflexion angle ($F_{(1, 13)}=9.975$, $p=0.008$, partial $\eta^2=0.434$), and peak plantar-flexion angle ($F_{(1, 13)}=5.287$, $p=0.039$, partial $\eta^2=0.289$; Table 2). Post hoc analyses revealed the following (Figure 4): (1) The ankle angle at foot strike in RG significantly decreased after training and was considerably less than that in CG after training. (2) The peak dorsiflexion angle in CG significantly increased after training and was considerably greater than that in RG after training. (3) The peak plantar-flexion angle in RG significantly increased after training. The significant main effect of time was observed for the range of motion ($F_{(1, 13)}=21.740$, $p<0.001$, partial $\eta^2=0.626$). The results indicated that the ankle's range of motion significantly increased.

For the morphological properties of AT, significant group-by-time interactions with large effect sizes were observed for the length ($F_{(1, 13)}=10.086$, $p=0.007$, partial $\eta^2=0.437$) and CSA ($F_{(1, 13)}=7.475$, $p=0.017$, partial $\eta^2=0.365$) of AT. Post hoc analyses revealed that the length and CSA of AT in RG increased after training.

For the mechanical properties of AT, a significant main effect of group was observed for the peak AT force ($F_{(1, 13)}=5.887$, $p=0.031$, partial $\eta^2=0.312$), peak AT elongation $F_{(1, 13)}=5.003$, $p=0.043$, partial $\eta^2=0.278$), and average AT loading rate ($F_{(1, 13)}=5.833$, $p=0.031$, partial $\eta^2=0.310$). The results revealed that peak AT force and peak AT elongation in RG were significantly greater than those in CG. However, the observed differences were not influenced by the intervention in the RG group. There were nonsignificant interactions for group-by-time indicated changes observed are not due to the intervention but rather reflect preexisting differences between the groups. A significant main effect of time was observed for the time-to-peak AT force ($F_{(1, 13)}=5.225$, $p=0.040$, partial $\eta^2=0.287$), average ($F_{(1, 13)}=7.228$, $p=0.019$, partial $\eta^2=0.357$), and peak AT loading rate ($F_{(1, 13)}=11.687$, $p=0.005$, partial $\eta^2=0.473$), but nonsignificant interaction for group-by-time was observed. The results revealed the time-to-peak AT force and the average and peak loading rates were increased after the 12-week training in both groups rather than the intervention having any effects (Table 3).

4 | DISCUSSION

All the participants in this study used to wear cushioned running shoes with RFS before training. After 12 weeks of training, eight of the participants in RG were running with FFS when shod, and ankle angle at initial contact changed from dorsiflexion before training to plantar flexion, which was consistent with the hypothesis of this study that the majority of runners would transition to FFS. These results were similar to those of McCarthy et al.,[15] who found that 56% (5/9) of female runners in their experimental group ran with a non-RFS pattern when shod after 12 weeks of simulated barefoot running, along with a more plantar-flexed angle at foot strike. Participants in their study were free to develop their own running pattern during the intervention and were not instructed on forefoot strike patterns. This finding was also consistent with the ankle kinematics of habitually FFS runners when shod running.[26,27] However, some runners in RG did not adopt an FFS pattern after training. This result could be attributed to the footwear used in the experiment, which involved a preferred strike pattern. McCarthy et al.[15] reported that 100% of their participants adopted non-RFS while wearing minimalist shoes, and 56% still maintained non-RFS while wearing cushioned shoes after their intervention.

TABLE 2 Effects of the 12-week gait retraining on cadence, strike angle, ankle joint kinematics, and kinetics during running (mean ± SD).

	Retraining group ($n=14$)		Control group ($n=14$)		p Value		
Variable	**Pre**	**Post**	**Pre**	**Post**	**Time**	**Group**	**Interaction**
Cadence (steps/min)	174.71 ± 8.30	176.22 ± 15.02	171.73 ± 14.87	169.03 ± 13.42	0.708	0.293	0.183
Strike angle (°)	12.81 ± 3.87	0.72 ± 7.77*	13.53 ± 4.14	14.48 ± 5.33#	<0.001	<0.001	<0.001
Ankle angle at foot strike (°)	5.31 ± 5.05	−4.08 ± 6.22*	5.83 ± 3.25	7.84 ± 4.41#	0.007	<0.001	0.002
Peak DF angle (°)	15.14 ± 3.49	12.98 ± 5.18	13.23 ± 3.46	17.9 ± 3.37*#	0.210	0.197	0.008
Peak PF angle (°)	12.61 ± 5.97	19.87 ± 5.80*	14.20 ± 6.96	15.42 ± 9.57	0.003	0.497	0.039
Ankle ROM (°)	27.75 ± 4.64	32.86 ± 4.22	27.43 ± 7.43	33.11 ± 9.33	<0.001	0.986	0.755
Peak ankle PF torque (Nm/kg)	2.77 ± 0.31	3.10 ± 0.46	2.89 ± 0.35	2.86 ± 0.35	0.109	0.547	0.083

Abbreviations: DF, dorsiflexion; PF, plantar flexion; ROM, range of motion.

*Significant difference before and after training in the same group, $p<0.05$.

#Significant difference between the two groups at the same test time, $p<0.05$.

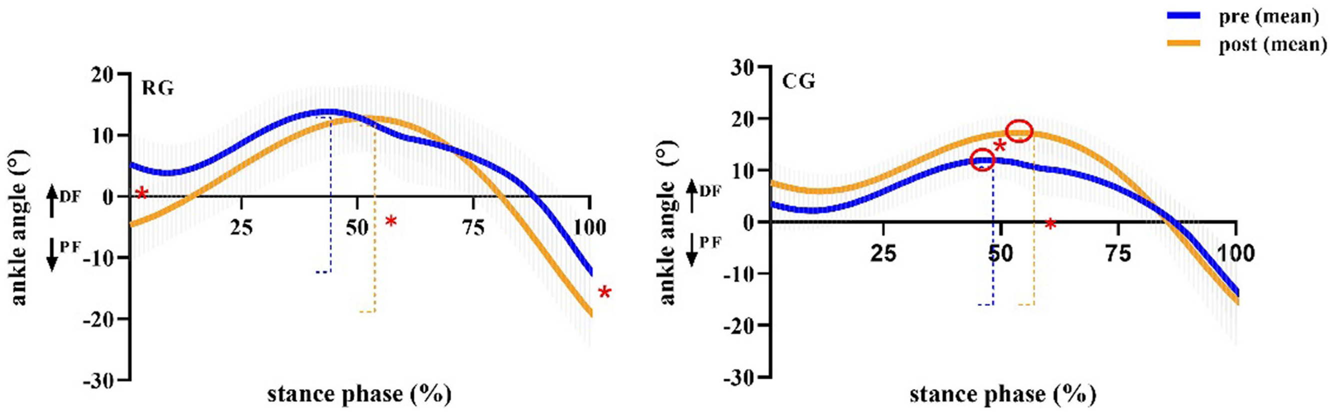

FIGURE 4 Effects of the 12-week gait retraining on ankle joint kinematics during the stance phase. Gray error bars show the standard error of the mean. The dashed line represented the difference between the maximum and minimum values, i.e., ankle joint range of motion. CG, control group; DF, dorsiflexion; PF, plantar flexion; RG, gait retraining group; *$p < 0.05$: significant difference before and after training.

Similarly, Lieberman et al.[28] reported that 50% of habitually barefoot runners maintained non-RFS during shod running. The results of the current study found that strike angle was decreased compared with that before training in the remaining members of RG who did not completely become FFS runners. Therefore, the runners who were accustomed to RFS could gradually transition to non-RFS or even FFS after 12 weeks of training.

Our findings supported the hypothesis that RG exhibited a significant 7.6% increase in AT CSA and a 4.9% increase in AT length at rest compared with CG after 12 weeks of training. A larger AT CSA supported the findings of Joseph et al.,[20] who found that the AT CSA of male runners increased by 3.4% after 12 weeks of training. During typical eccentric activities, the closer the AT approached the safety limit, the greater the tendon strains and stresses produced. In the present study, it can be observed that despite a 13% increase in AT forces, there were no differences in tendon stress attributed to the greater cross-sectional area of the tendon after training. Thereby, a larger AT cross-sectional area may contribute to a greater “safety factor” against tendon injury, as supported by the study of Rosager et al.[29] A longer AT length in RG was proven to be beneficial for improving running performance among endurance runners.[4] Similarly, Hunter et al.[30] determined that longer AT exhibited greater potential to store elastic energy with better running economy. AT length has been shown to be independently associated with stretch-shortening cycle potentiation (SSCP) force and SSCP velocity. A longer AT could enhance the stretching and velocity potential of the muscle–tendon complex while running.[30] This observation was consistent with the AT structure of elite runners, whose longer AT was advantageous for middle- and long-distance performance. Therefore, our findings suggested that longer and thicker AT might be a beneficial training adaptation for long-distance runners. Previous studies that involved 12-week eccentric training or 14-week plyometric training reported that no significant changes were found in AT CSA.[31,32] Furthermore, the results of the aforementioned studies suggested that 12 weeks of gait retraining might be more effective in changing AT's static morphological characteristics than resistance training.

Theoretically, increased ankle plantar flexion is required at initial contact in FFS pattern, which may require triceps strength to control the heel drop and propel the body forward. Minimalist shoes with an FFS pattern were associated with a loading stimulus that was likely to induce AT adaptation through a repetitive elongation–contraction cycle. However, in contrast to our hypothesis, the current study did not find a group-by-time interaction for AT force. This is inconsistent with the study of Joseph et al.[20] who observed an increase in AT force after 12-week training with minimalist shoes, although AT force was measured by maximum isometric contraction on the isokinetic plyometric tester, which differed from the present study which calculated AT force during running. The current findings were also inconsistent with greater AT forces in habitual forefoot strikers than rearfoot strikers.[33,34] The evidence of the influence of exercise on the mechanical properties of tendons is inconclusive in the longitudinal intervention studies. Hansen et al.[35] observed that the load-deformation curve of the high-stress tendon-aponeurosis complex of the triceps surae did not produce a measurable change after untrained human subjects started running for a prolonged time, which was consistent with the current study. Therefore, we speculated that it might be necessary to observe significant changes in AT loading over a longer training period.

The results showed nonsignificant group-by-time interaction for the peak AT elongation and strain, namely no significant difference was noted in peak AT elongation

TABLE 3 Effect of the 12-week gait retraining on AT loading during running (mean ± SD).

	Retraining group ($n = 14$)			Control group ($n = 14$)			p Value		
Variable	**Pre**	**Post**	**Percent change (%)**	**Pre**	**Post**	**Percent change (%)**	**Time**	**Group**	**Interaction**
AT length (cm)	19.81 ± 3.05	20.72 ± 2.85*	4.9	20.53 ± 1.92	20.61 ± 1.83	0.5	0.001	0.970	0.007
AT CSA (mm^2)	61.28 ± 5.64	65.86 ± 6.28*	7.6	59.59 ± 7.41	60.79 ± 7.46	2.1	<0.001	0.373	0.017
Moment arm of AT (cm)	4.97 ± 0.34	4.93 ± 0.14	−0.6	5.06 ± 0.18	4.95 ± 0.14	−2.1	0.118	0.259	0.519
Peak AT force (BW)	5.98 ± 0.95	6.60 ± 1.10	13.3	5.79 ± 0.71	5.88 ± 0.74	2.3	0.096	0.031	0.325
Time-to-peak AT force (ms)	126.49 ± 16.11	117.29 ± 10.74	−6.2	130.36 ± 17.56	129.04 ± 11.75	−0.04	0.040	0.153	0.227
Average AT LR (BW/s)	47.95 ± 8.75	57.26 ± 12.07	22.6	45.44 ± 9.63	46.17 ± 7.42	4.0	0.019	0.031	0.127
Peak AT LR (BW/s)	255.48 ± 93.56	312.26 ± 90.40	33.8	223.53 ± 64.16	292.36 ± 68.79	37.2	0.005	0.306	0.766
AT force impulse (BW·s)	0.67 ± 0.11	0.74 ± 0.15	14.1	0.66 ± 0.09	0.67 ± 0.11	1.8	0.189	0.136	0.216
Peak AT stress (MPa)	69.11 ± 9.34	69.67 ± 15.58	1.7	64.26 ± 11.06	64.21 ± 11.42	0.5	0.912	0.221	0.901
Peak AT elongation (mm)	7.22 ± 1.31	7.77 ± 1.90	11.1	7.13 ± 1.63	7.24 ± 1.46	7.8	0.464	0.022	0.627
Peak AT strain (%)	3.75 ± 0.99	3.82 ± 1.08	10.3	3.50 ± 0.90	3.34 ± 1.26	3.1	0.862	0.152	0.701
AT work (J)	6.76 ± 3.40	6.76 ± 3.14	8.4	6.74 ± 2.59	6.66 ± 1.55	14.4	0.944	0.952	0.942

Abbreviations: AT, Achilles tendon; CSA, cross-sectional area; LR, loading rate.

*Significant difference before and after training in the same group, $p < 0.05$.

and strain after training. Similarly, Suzuki et al. reported no significant effect of running speed (10 km/h versus 14 km/h versus 18 km/h) and strike pattern (FFS versus RFS) on the elongation of AT during running. In addition, the elongation caused by AT force was suggested to be important for the elastic storage and recovery in AT during running.[36] Increased AT force and elongation during running could convert a higher proportion of kinetic energy into elastic strain energy.[37] On the basis of the musculoskeletal simulation of the plantar-flexion muscular–tendon model, Yong et al.[38] found that the energy storage of AT and the work done by the medial gastrocnemius muscle fiber could increase with an immediate FFS pattern. By contrast, the results of the current study found no significant differences in AT work (energy release) at 10 km/h before and after the 12-week gait retraining, indicating that the elastic energy contribution of AT was relatively constant before and after training. The current study observed that the differences in AT length in RG before and after training primarily occurred at initial contact, while the release of elastic energy of AT mostly occurred in the middle and late stance phases. We assumed that similar changes in AT length before and after training during the stance phase might not affect the release of elastic energy of AT. Suzuki et al.[19] reported a similar finding that the length of AT from the time of its peak elongation to toe-off appeared to be similar between FFS and RFS via direct measurement. Furthermore, few studies have directly measured changes in the length of the entire AT during running by using real-time ultrasound imaging, and thus, this technique must be explored further in future studies.

Recent studies have generally concurred that AT is sensitive to changes in the mechanical environment. In accordance with the load-induced tendinopathy model, degenerative tendinopathy is associated with chronic excessive load, and weak AT occurs when unloading.[39] That is, only stimulation within the optimal mechanical loading range can positively promote AT homeostasis, resulting in adaptive changes in tendon tissues.[40] In vitro studies of AT have shown that failure stress and strain are 100 MPa and 10%, respectively.[41,42] The results of the current study found that AT stress (mean values were 69.7 MPa in RG and 64.2 MPa in CG after training) and strain (mean values were 3.8% in RG and 3.3% in CG after training) were lower than the failure stress and strain in in vitro studies, preliminarily suggesting that the gradual transition to FFS could improve AT load safely. However, the optimal range of loading on AT in vivo remains unclear at present due to many factors, e.g., strain rate and strain duration. Therefore, further research is necessary to investigate the specific loading range for positive tendon adaptation in the future.

The current study has certain limitations. Firstly, despite the utilization of ultrasonography in our study to assess AT morphology, it's important to acknowledge that this approach has limitations in terms of image resolution compared to MRI. This choice was driven by factors such as accessibility, cost, and real-time monitoring potential, even though it might have led to reduced visualization of certain structural details. Secondly, although all the participants were of similar ages and running experiences, individual differences may occur in the feedback of AT when running in FFS with five-fingered shoes. Furthermore, the assessment of AT elongation did not consider the curvature of the tendon identified by the markers placement due to the limitations imposed by footwear conditions. Therefore, admittedly, AT mechanical properties during running might be underestimated when its curvature is not accounted for in this study.

5 | CONCLUSION

The findings of this study provided real-time feedback on AT based on ultrasound images. Preliminary evidence was found that a gradual transition program from RFS to FFS could facilitate favorable morphological changes in the AT, i.e., increased CSA and length. However, no differences in the mechanical properties of AT were observed. It can therefore be suggested the 12-week training conducted in this study was insufficient to provoke mechanical loading changes on AT. This finding will serve as a basis for future studies to explore the appropriate dose response that will maximize AT adaptation and minimize injury risk.

6 | PERSPECTIVES

According to current clinical concepts, the exercise that applies mechanical loading to the AT is supported by the highest level of evidence as a treatment option. Given the differences in the AT of runners with different foot strike pattern, this study investigated on the effects of a gradual transition program from an RFS to an FFS on AT. The use of real-time ultrasound imaging provided valuable feedback on the tendon's response to the gait retraining program. It was observed that such retraining positively influenced the morphological properties of AT, i.e., increased CSA and length, enhancing against injury and stretching potential while running. Gait retraining programs hold promise for individuals seeking to improve their running mechanics. These preliminary results serve as an important foundation for future studies aiming to determine the appropriate dosage response required to maximize AT adaptation while minimizing the risk of injury. By establishing an optimal balance between the intensity and duration of the gait retraining program, researchers can further explore how to promote the desired morphological and mechanical adaptations of the tendon. Additionally, investigating the long-term effects of gait retraining on the AT morphology will provide valuable insights into the sustainability and durability of these adaptations.

AUTHOR CONTRIBUTIONS

ZXN wrote the first draft of the manuscript and analyzed and interpreted the data from the subjects. Methodological quality assessments were conducted by DLQ, XSL, and FWJ. FWJ contributed to idea conception. ZXN and DLQ conducted the data collection and supervised the training intervention. All authors read and approved the final manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

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